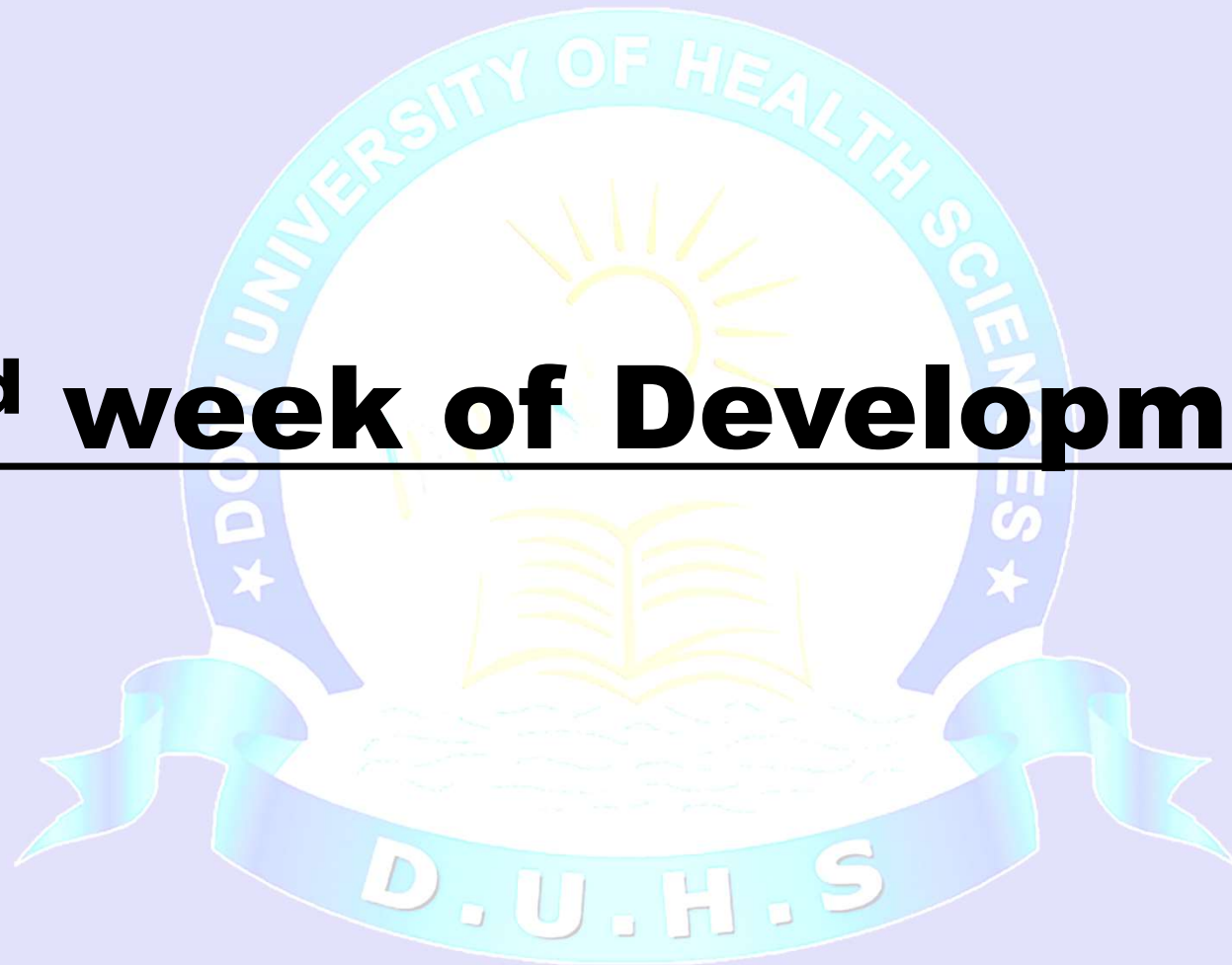


2nd week of Development.



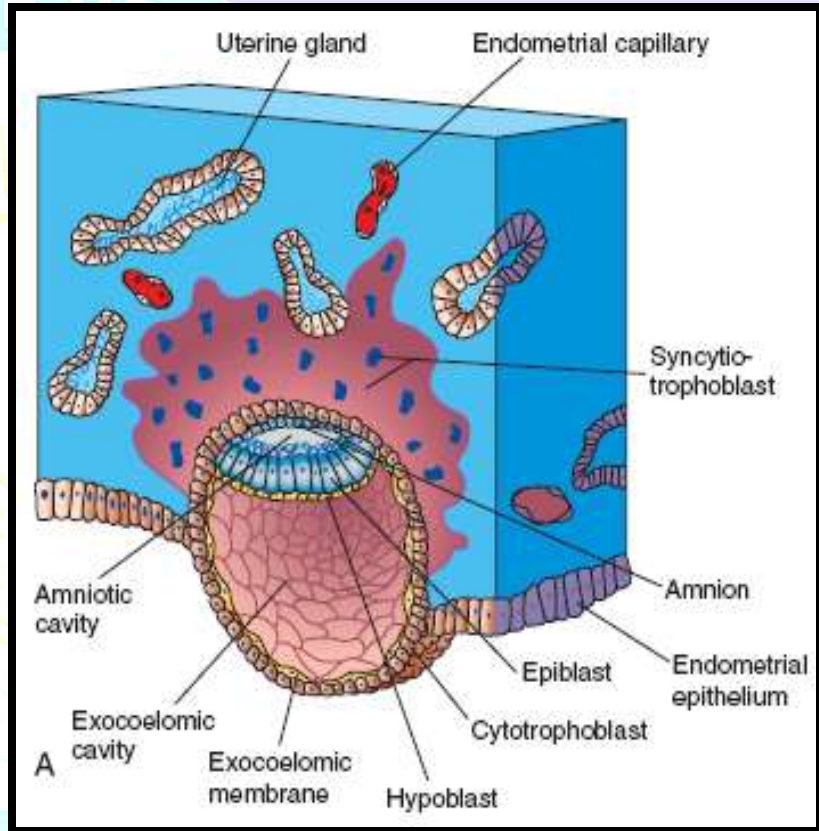
LEARNING OBJECTIVES

- At the end of the lecture, student should be able to
 - Define implantation
 - State its normal site
 - Explain the formation of outer and inner cell masses
 - Discuss the further development of outer cell mass (trophoblast),
 - Differentiate syncytiotrophoblast and cytotrophoblast with its microscopic appearance
 - Describe the process of implantation (day wise change)
 - State the differentiation of embryonic pole and development of bilaminar germ disc with formation Epiblast and hypoblast, their cavities (amniotic cavity and primary yolk sac)
 - Discuss the development of the chorionic sac and formation Primary chorionic villi
 - Enumerate the abnormal sites for implantation (ectopic pregnancy) and the different diagnostic tools

IMPLANTATION

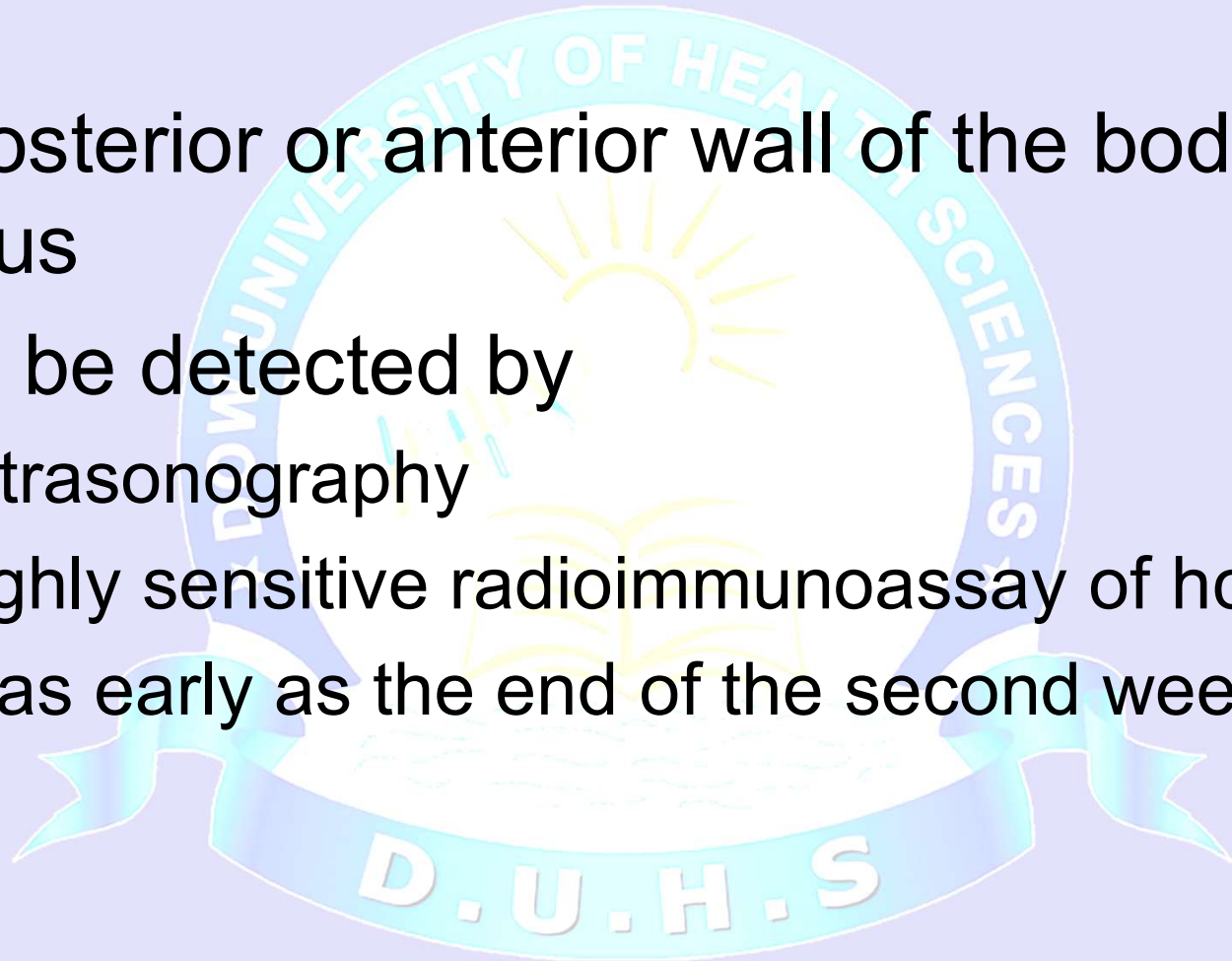
Definition:

- It is the attachment of blastocyst to the endometrial layer of the uterine wall.
- Implantation of the blastocyst is completed during the second week.
- Morphologic changes in the embryoblast produce a bilaminar embryonic disc
- Composed of:
 - epiblast
 - hypoblast .



COMMON SITE OF IMPLANTATION

- Is posterior or anterior wall of the body of uterus
- Can be detected by
 - Ultrasonography
 - highly sensitive radioimmunoassay of hcg (as early as the end of the second week).

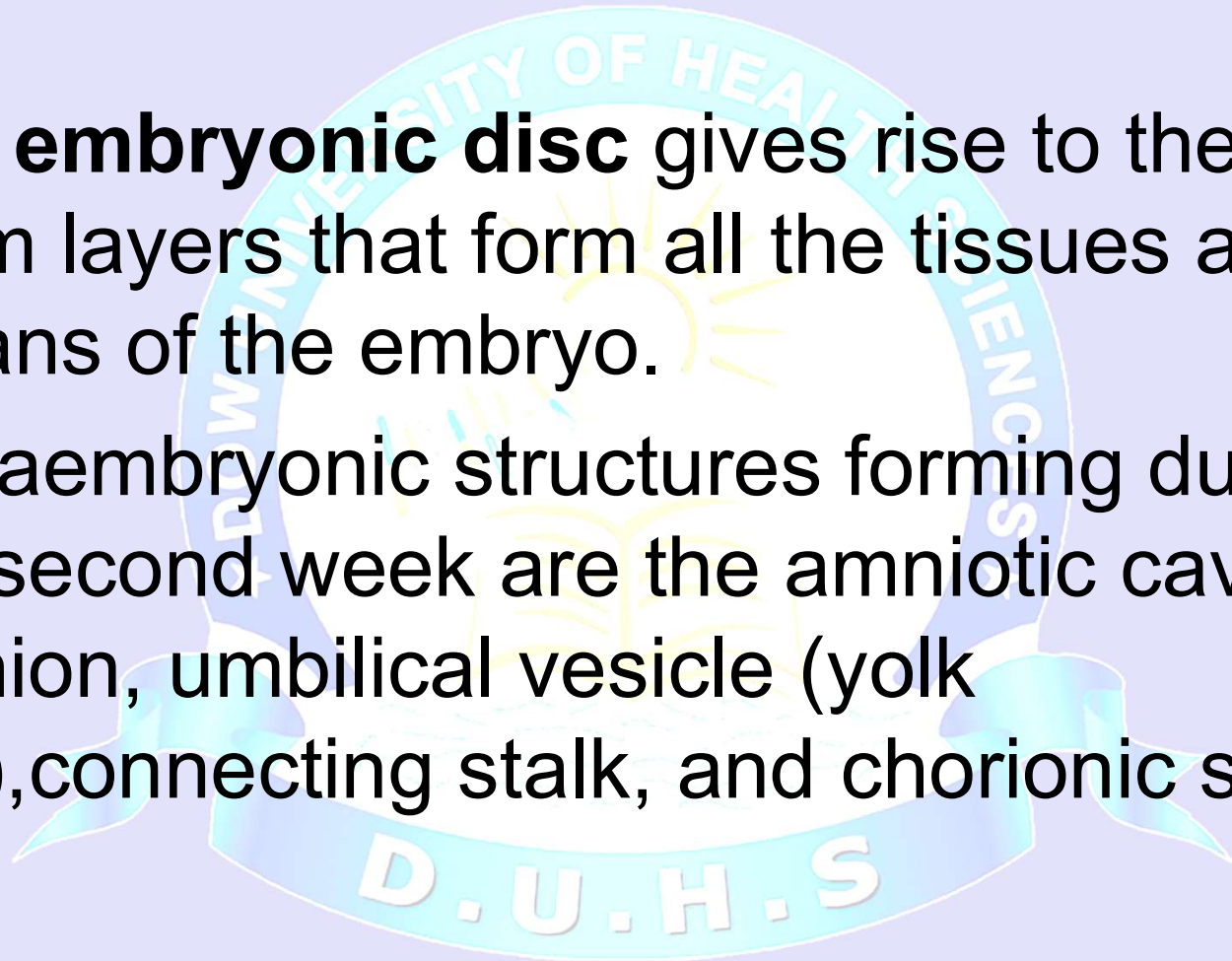


FORMATION OF OUTER & INNER CELL MASS

- **The inner cells mass** :Cells of the blastocyst differentiate into two layers
 1. **Layer of tall columnar cells** called epiblast adjacent to the amniotic cavity
 2. **Layer of low cuboidal** cells called hypoblast adjacent to the blastocyst cavity
- **Outer cell mass** :also differentiate into two layers
 1. **Layer of mononucleated cells** called cytotrophblast. These cells are active mitotically
 2. **Layer of multinucleated zone** without distinct cell boundaries called syncytotrophblast or syncitium

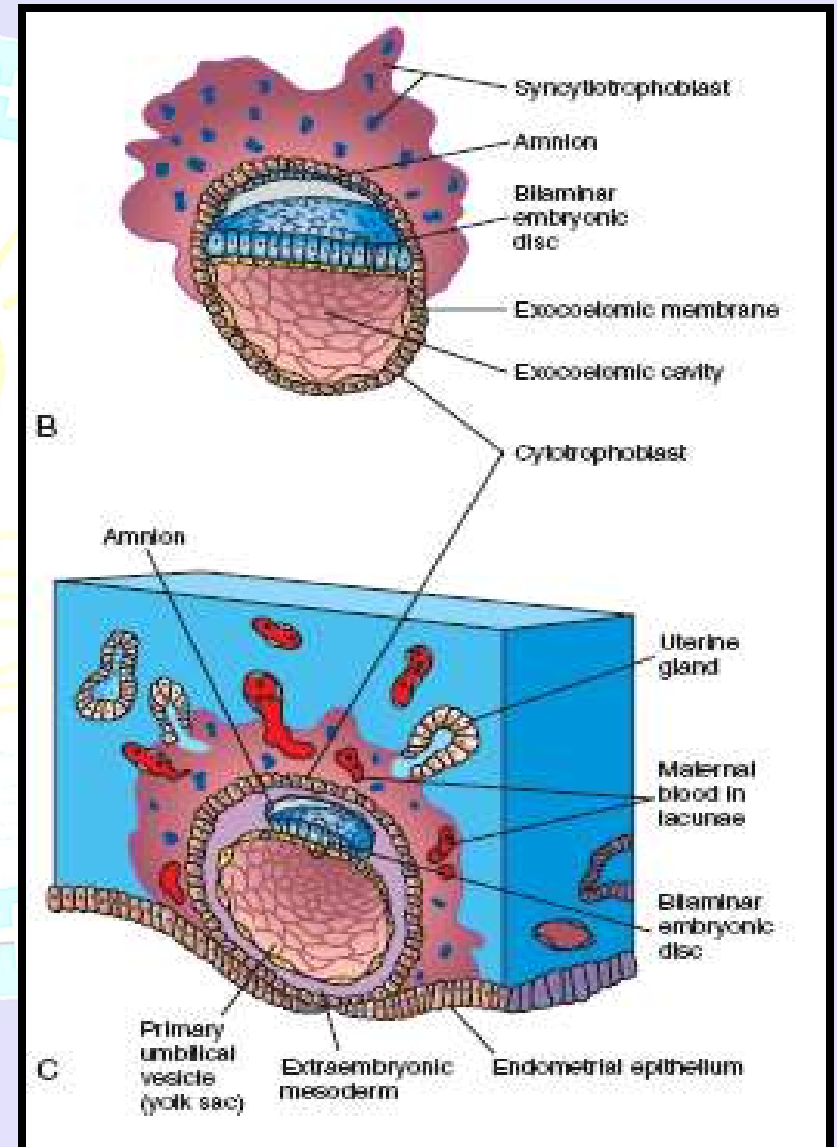
Extraembryonic structures

- The **embryonic disc** gives rise to the germ layers that form all the tissues and organs of the embryo.
- Extraembryonic structures forming during the second week are the amniotic cavity, amnion, umbilical vesicle (yolk sac), connecting stalk, and chorionic sac.



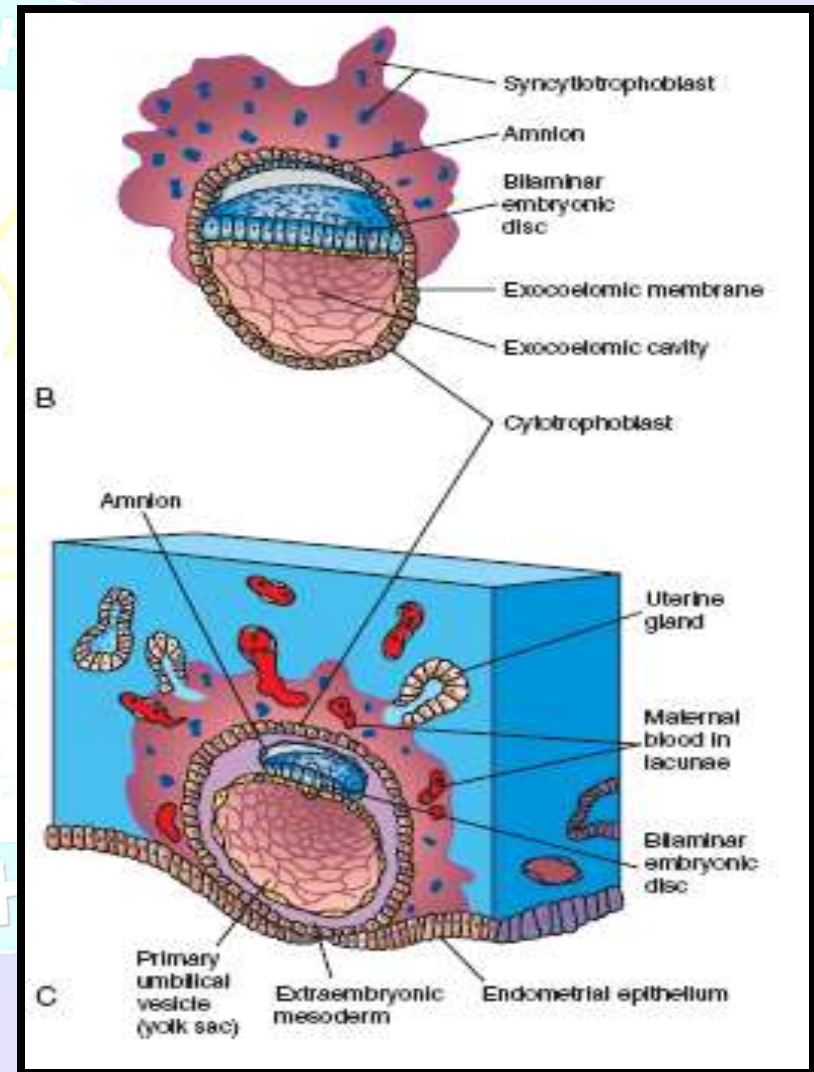
DEVELOPMENT OF TROPHOBLAST

- Implantation is completed by the end of the second week.
- It occurs during 6 to 10 days after ovulation.
- **Trophoblast differentiates into:**
 - **The cytotrophoblast:**
 - forms new cells that migrate into syncytiotrophoblast.
 - **The syncytiotrophoblast:**
 - Multinucleated mass in which no cell boundaries are discernible.



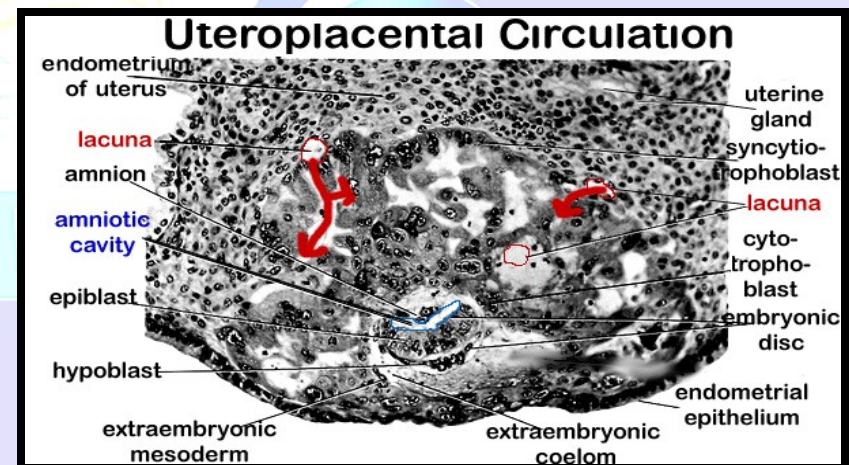
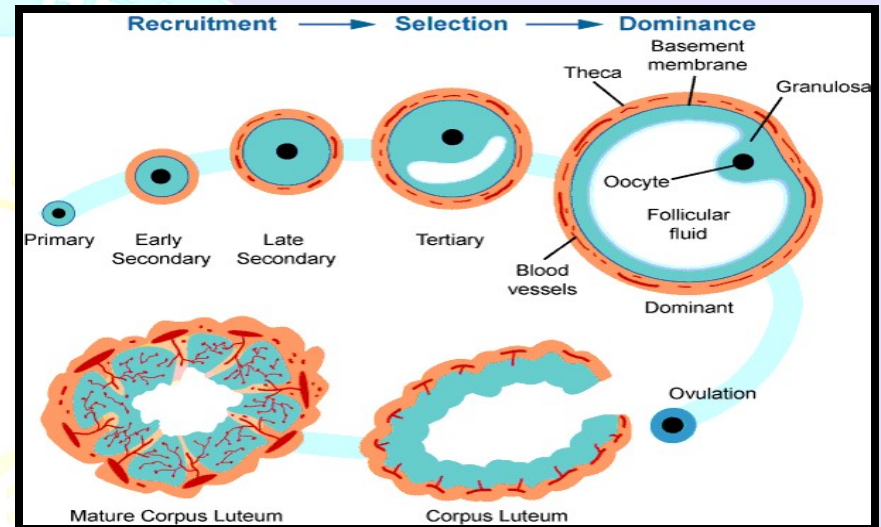
DEVELOPMENT OF TROPHOBLAST

- **syncytiotrophoblast** invades the endometrial connective tissue, and the blastocyst slowly embeds itself in the endometrium.
- Syncytiotrophoblastic cells displace endometrial cells at the implantation site.
- endometrial cells undergo **apoptosis**, which facilitates the invasion.



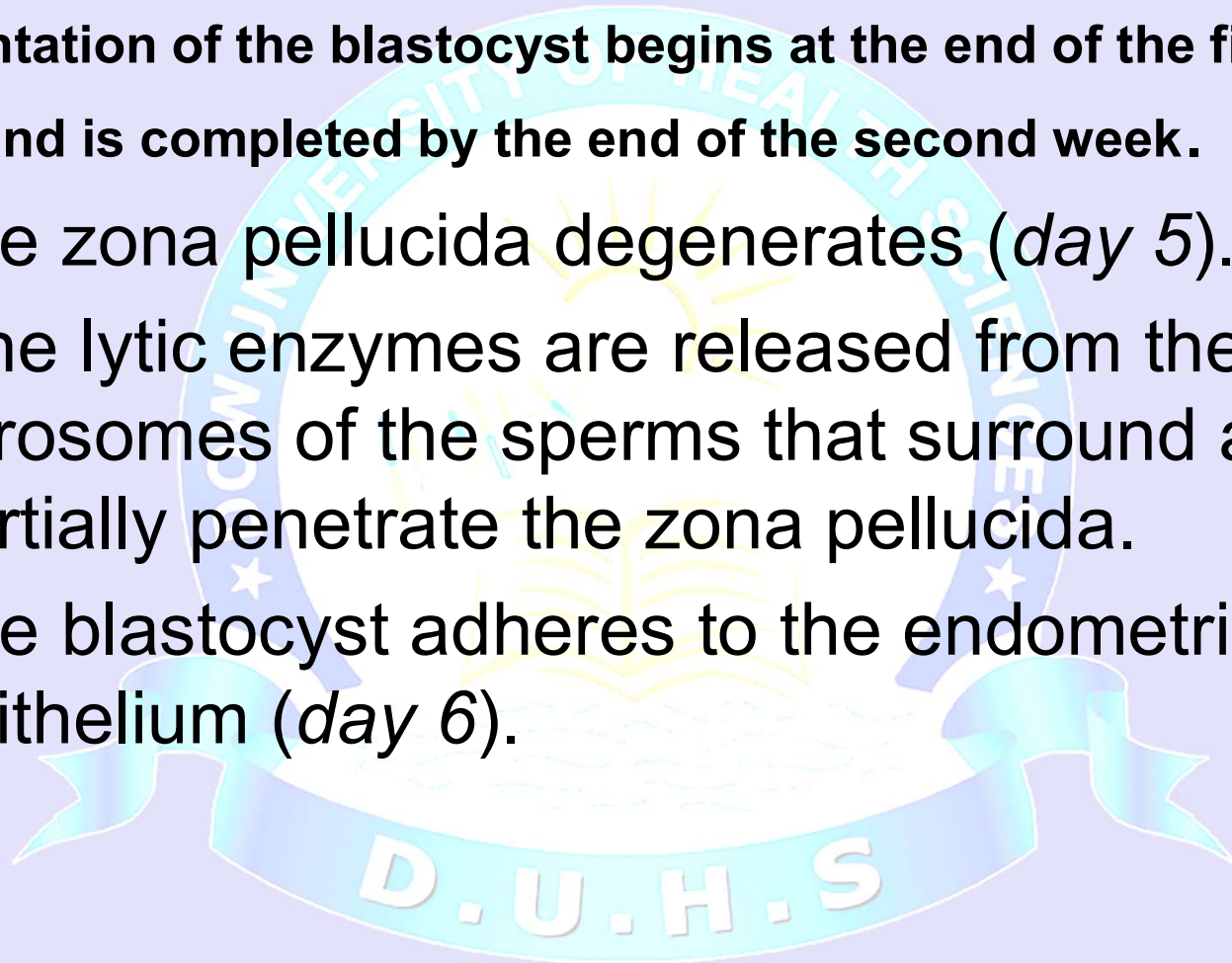
UTERO-PLACENTAL CIRCULATION

- **VACUOLES** appear within the syncytiotrophoblast which collapse to form lacunae (lacunar stage)
- syncytiotrophoblast penetrate deeper into the maternal endometrium & erode the endothelial wall of the maternal capillaries(sinusoids) & due to this maternal blood flow into the lacunar system & this starts the utro- placental circulation,
- The corpus luteum is an endocrine glandular structure that secretes estrogen and progesterone to maintain the pregnancy.

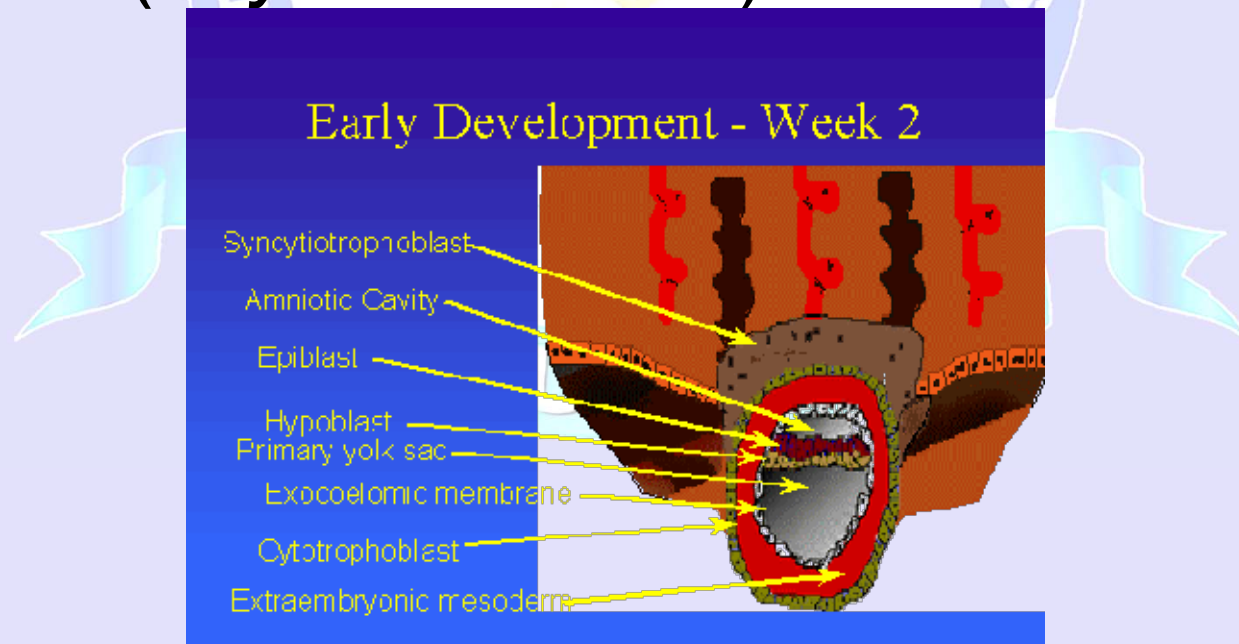


SUMMARY OF IMPLANTATION

- **Implantation of the blastocyst begins at the end of the first week and is completed by the end of the second week.**
 - The zona pellucida degenerates (*day 5*).
 - The lytic enzymes are released from the acrosomes of the sperms that surround and partially penetrate the zona pellucida.
 - The blastocyst adheres to the endometrial epithelium (*day 6*).



- The trophoblast differentiates into two layers:
 - The syncytiotrophoblast and
 - cytotrophoblast (*day 7*).
- Blood-filled lacunae appear in the syncytiotrophoblast (*day 9*).
- Lacunar networks form by fusion of adjacent lacunae (*days 10 and 11*).



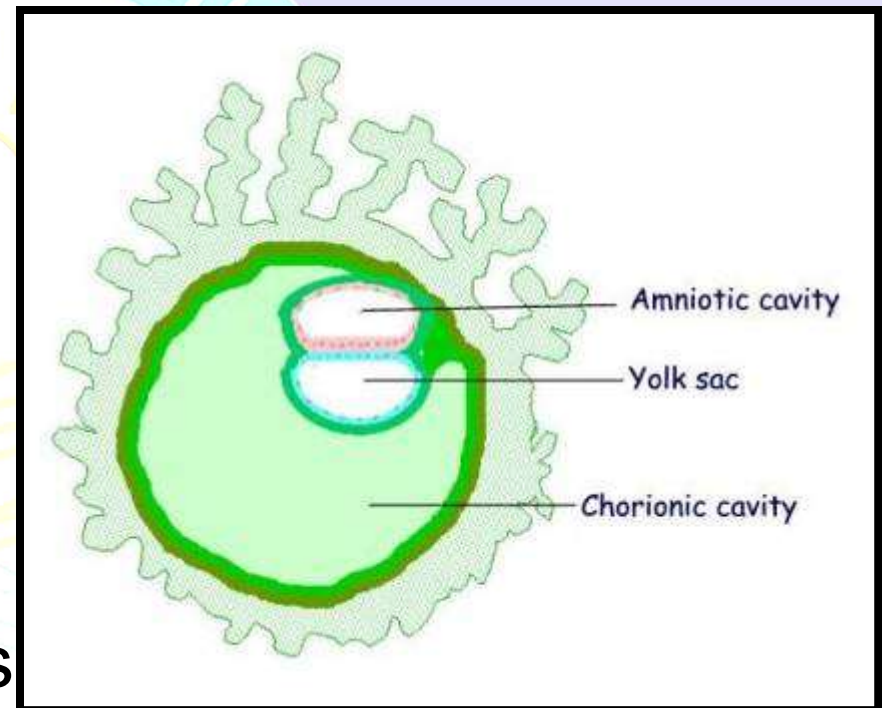
- The syncytiotrophoblast erodes endometrial blood vessels, allowing maternal blood to seep in and out of lacunar networks, thereby establishing a uteroplacental circulation (*days 11 and 12*).
- The defect in the endometrial epithelium is repaired (*days 12 and 13*).
- Primary chorionic villi develop (*days 13 and 14*).

NUTRITIVE SUBSTANCES

- *Oxygenated blood* passes into the lacunae from the *spiral endometrial arteries*, and *poorly oxygenated blood* is removed from them through the endometrial veins.
- *The 10-day embryo and Extraembryonic membranes* is completely embedded in the endometrium .

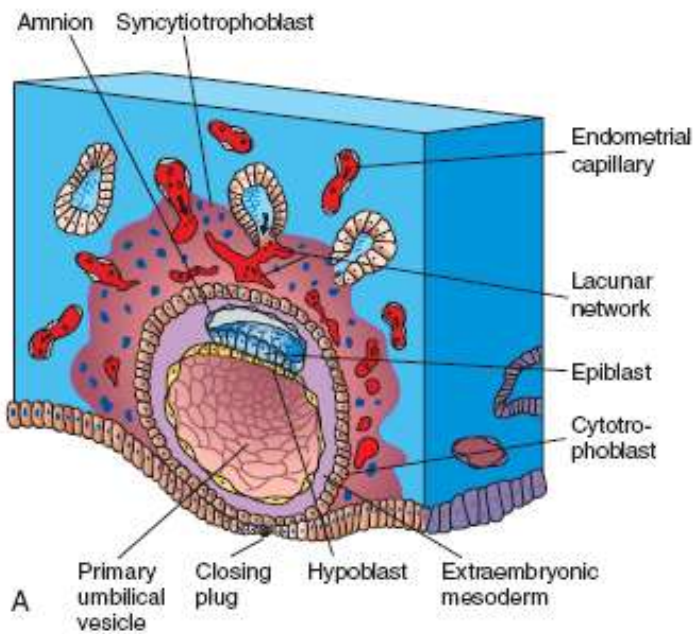
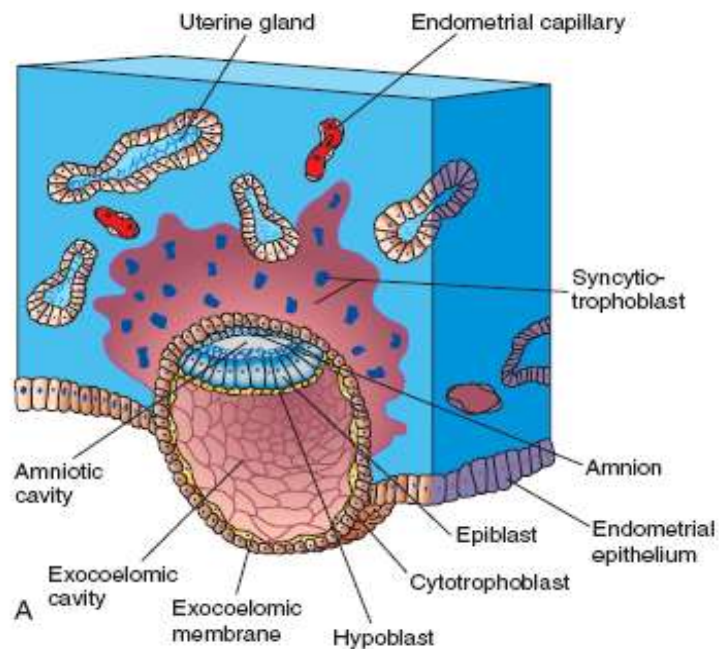
FORMATION OF THE AMNIOTIC CAVITY,

- As implantation of the blastocyst progresses, a small space "the primordium of the **amniotic cavity**" appears in the embryoblast.
- ***Amnioblasts***—separate from the epiblast and form the **amnion**, which encloses the amniotic cavity.

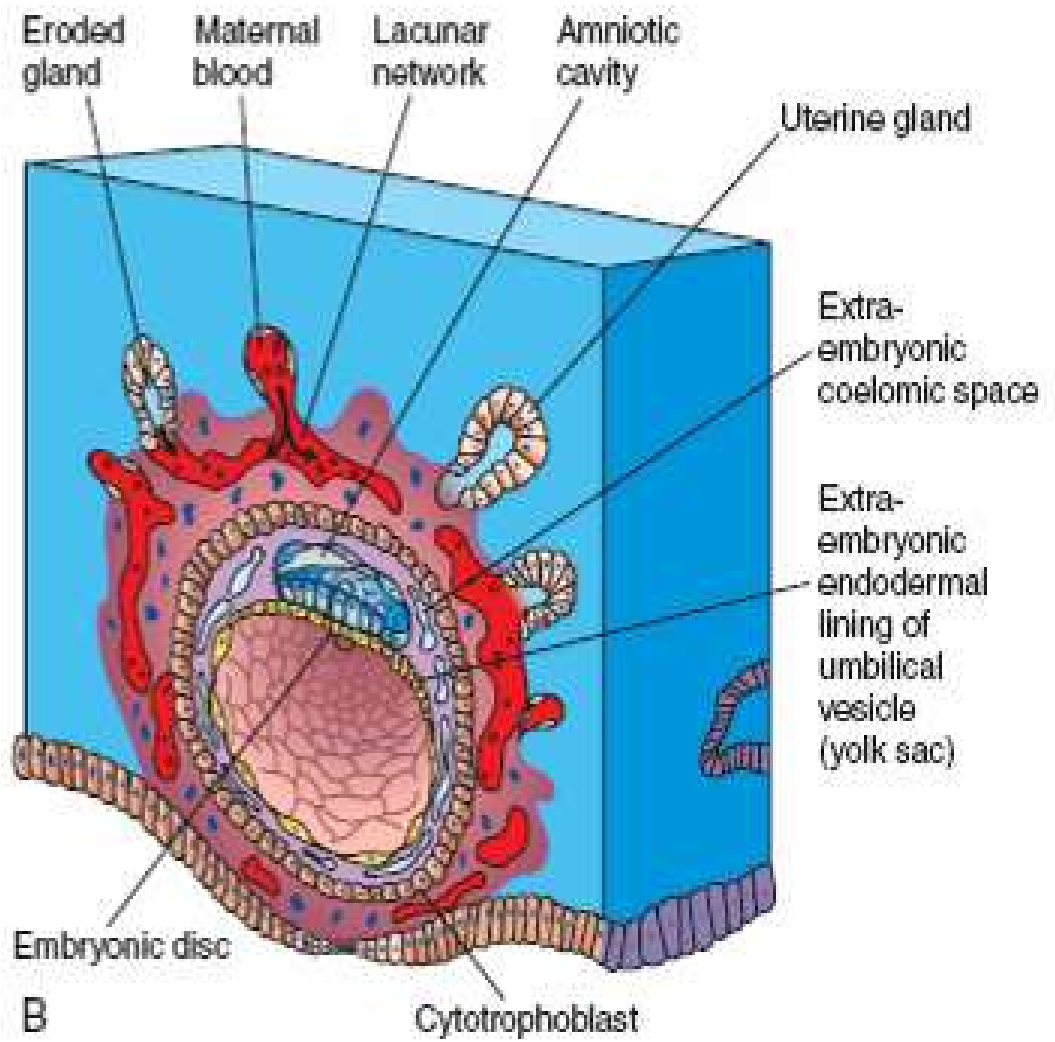


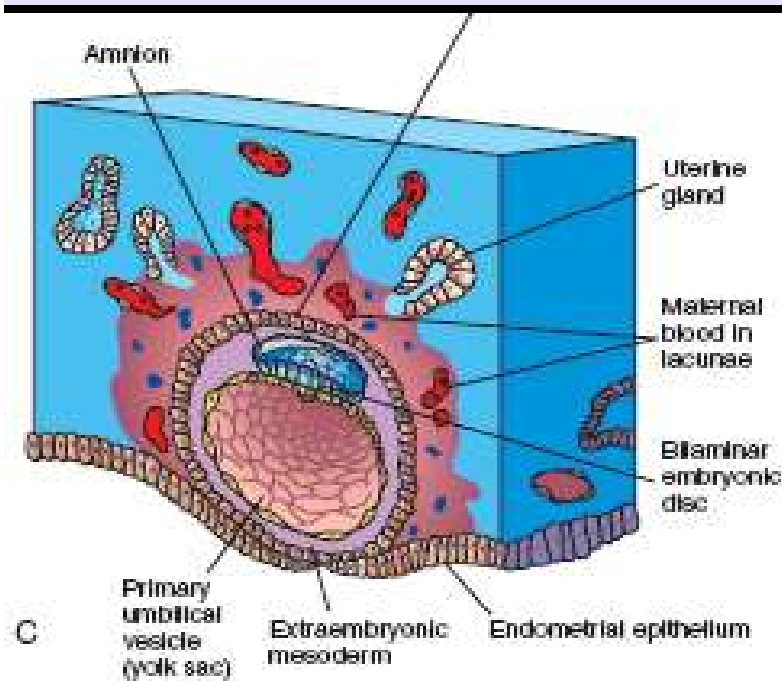
FORMATION OF EMBRYONIC DISC, AND UMBILICAL VESICLE

- Embryoblast results in the formation of **embryonic disc**, consisting of two layers:
 - **Epiblast :**
 - forms the floor of the amniotic cavity and is continuous peripherally with the amnion
 - **Hypoblast:**
 - forms the roof of the **exocoelomic cavity** and is continuous with the thin exocoelomic **membrane**.
 - This membrane, together with the hypoblast, lines the **primary umbilical vesicle**.



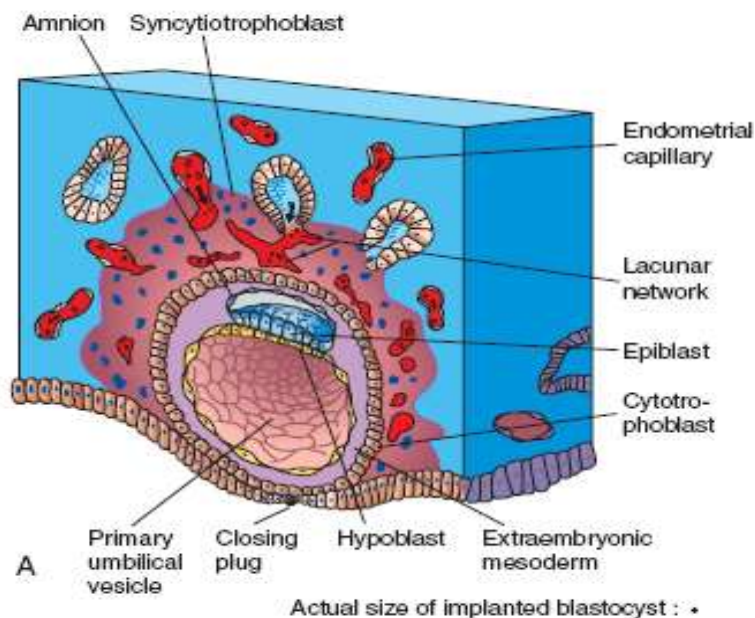
Actual size of implanted blastocyst : •





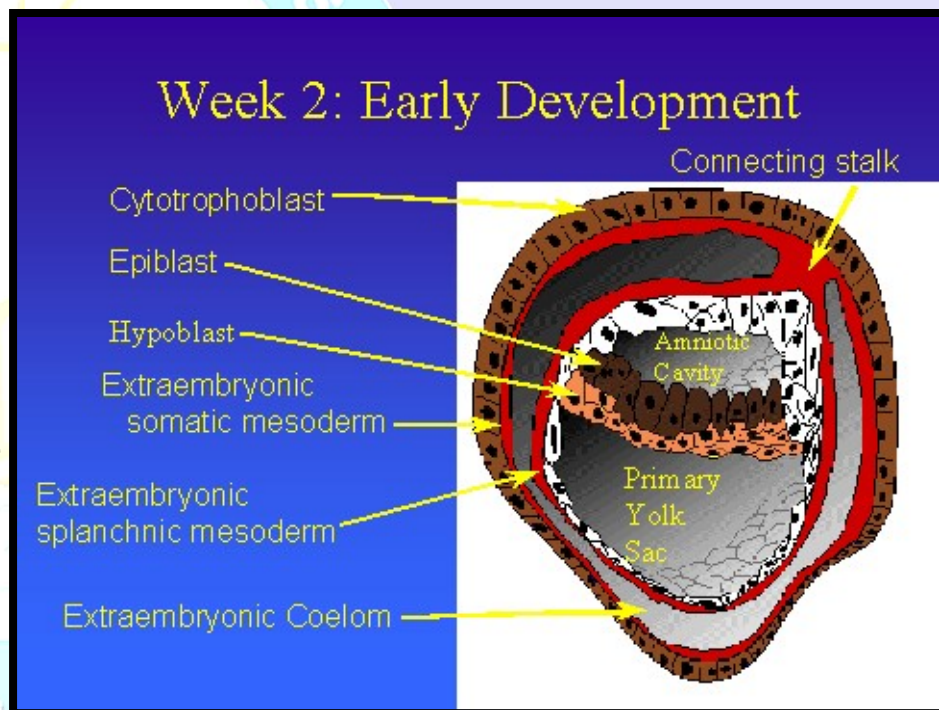
- The embryonic disc now lies between the amniotic cavity and the umbilical vesicle.

- Cells from the vesicle endoderm form a layer of connective tissue, the **extraembryonic mesoderm**.



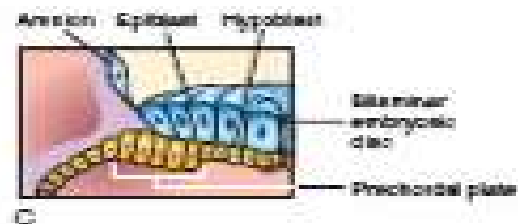
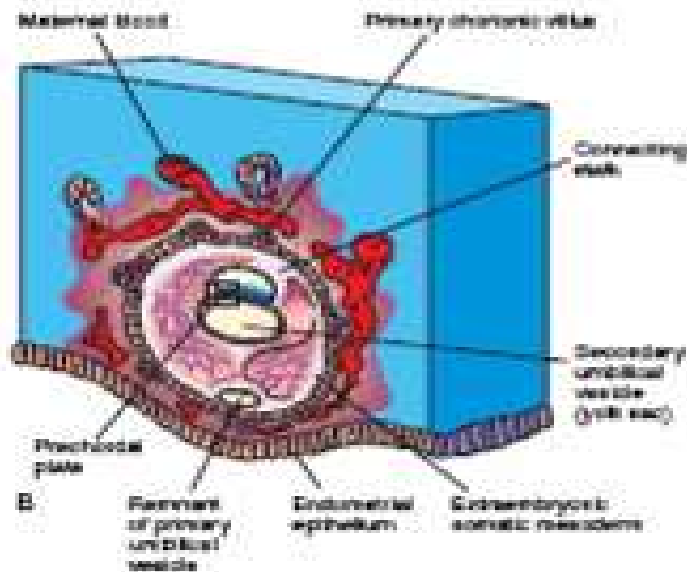
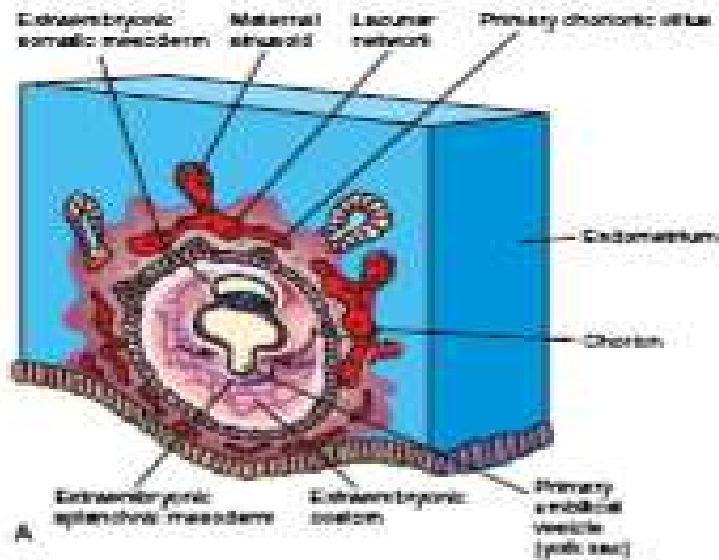
THE EXTRAEMBRYONIC COELOM

- The extraembryonic coelom splits the Extraembryonic mesoderm into two layers.
 - Extraembryonic somatic mesoderm, lining the trophoblast and covering the amnion
 - Extraembryonic splanchnic mesoderm, surrounding the umbilical vesicle



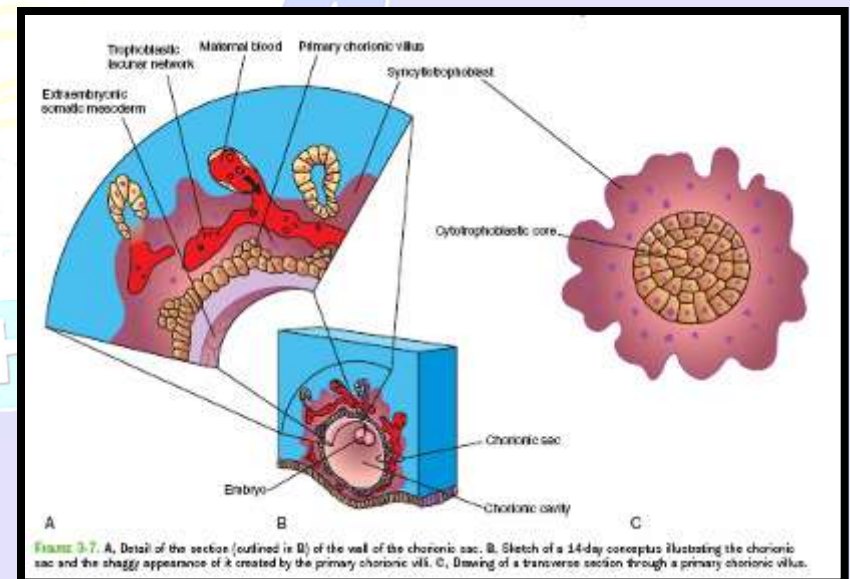
DEVELOPMENT OF CHORIONIC SAC

- **Primary chorionic villi appear at end of the second week.**
- Growth of cytotrophoblastic cellular extensions (finger like) is thought to be induced by the underlying **extraembryonic somatic mesoderm.**



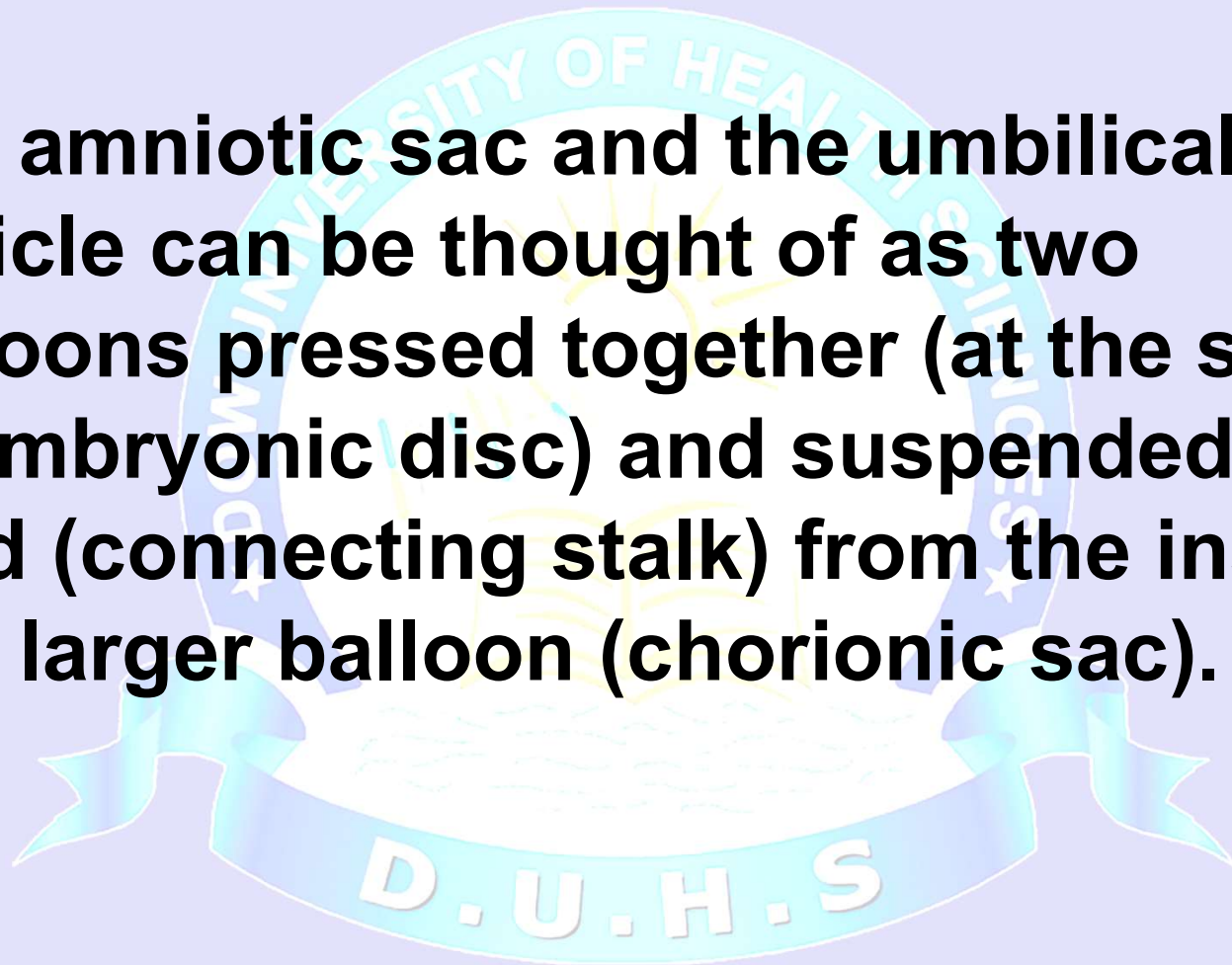
CHORION

- The extraembryonic somatic mesoderm and the two layers of trophoblast form the **chorion**.
- The *chorion forms the wall of the chorionic sac*, within which the embryo and its amniotic sac and umbilical vesicle are suspended by the connecting stalk.
- Extraembryonic coelom is now called the **chorionic cavity**.



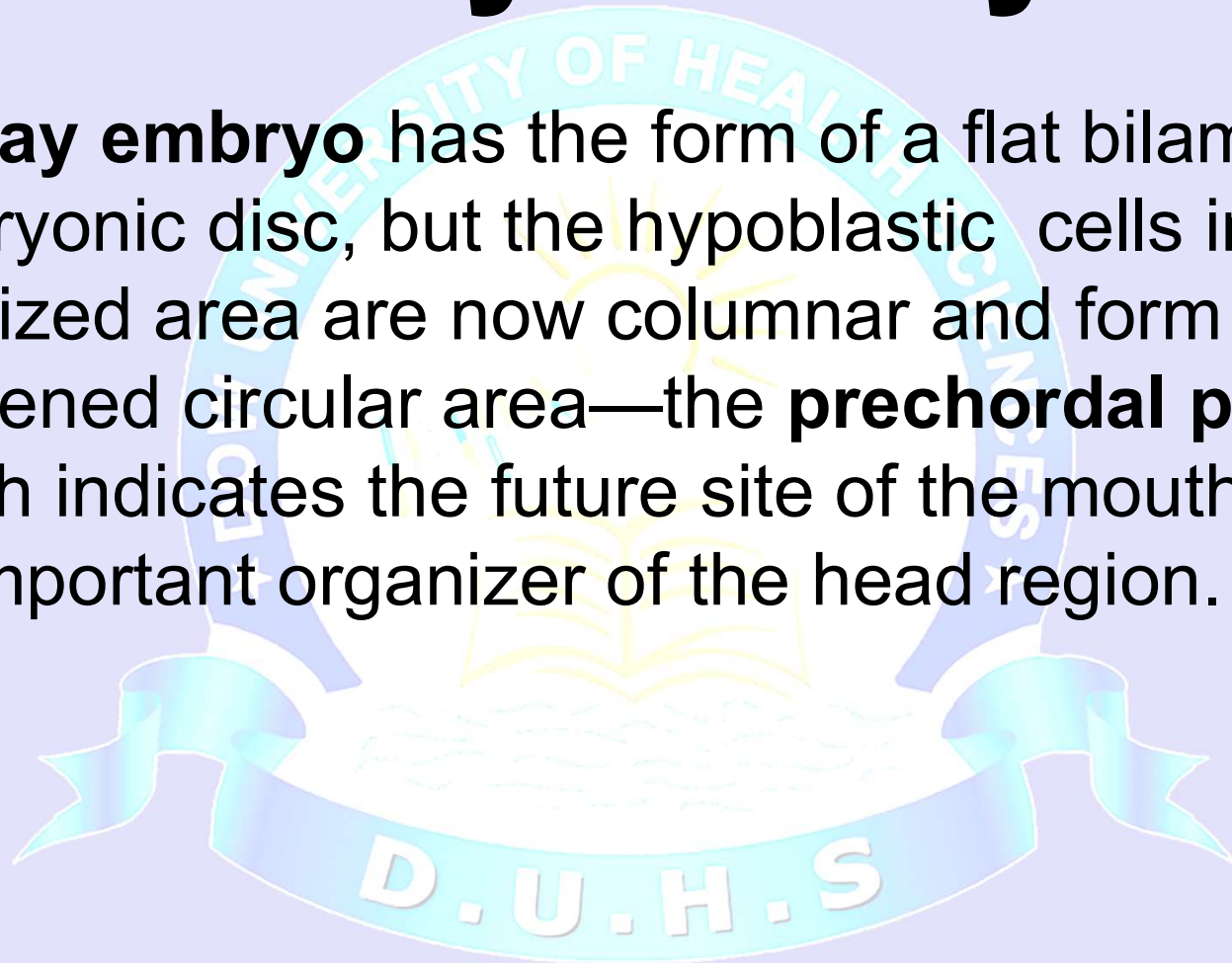
Amniotic sac

- **The amniotic sac and the umbilical vesicle can be thought of as two balloons pressed together (at the site of embryonic disc) and suspended by a cord (connecting stalk) from the inside of a larger balloon (chorionic sac).**



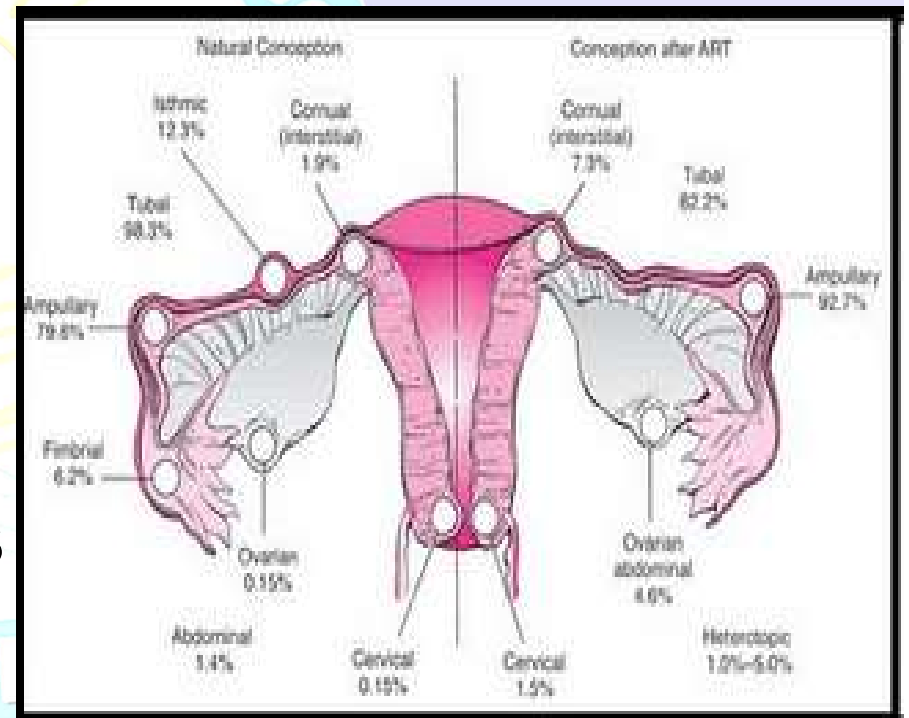
14-day embryo

- **14-day embryo** has the form of a flat bilaminar embryonic disc, but the hypoblastic cells in a localized area are now columnar and form a thickened circular area—the **prechordal plate** which indicates the future site of the mouth and an important organizer of the head region.



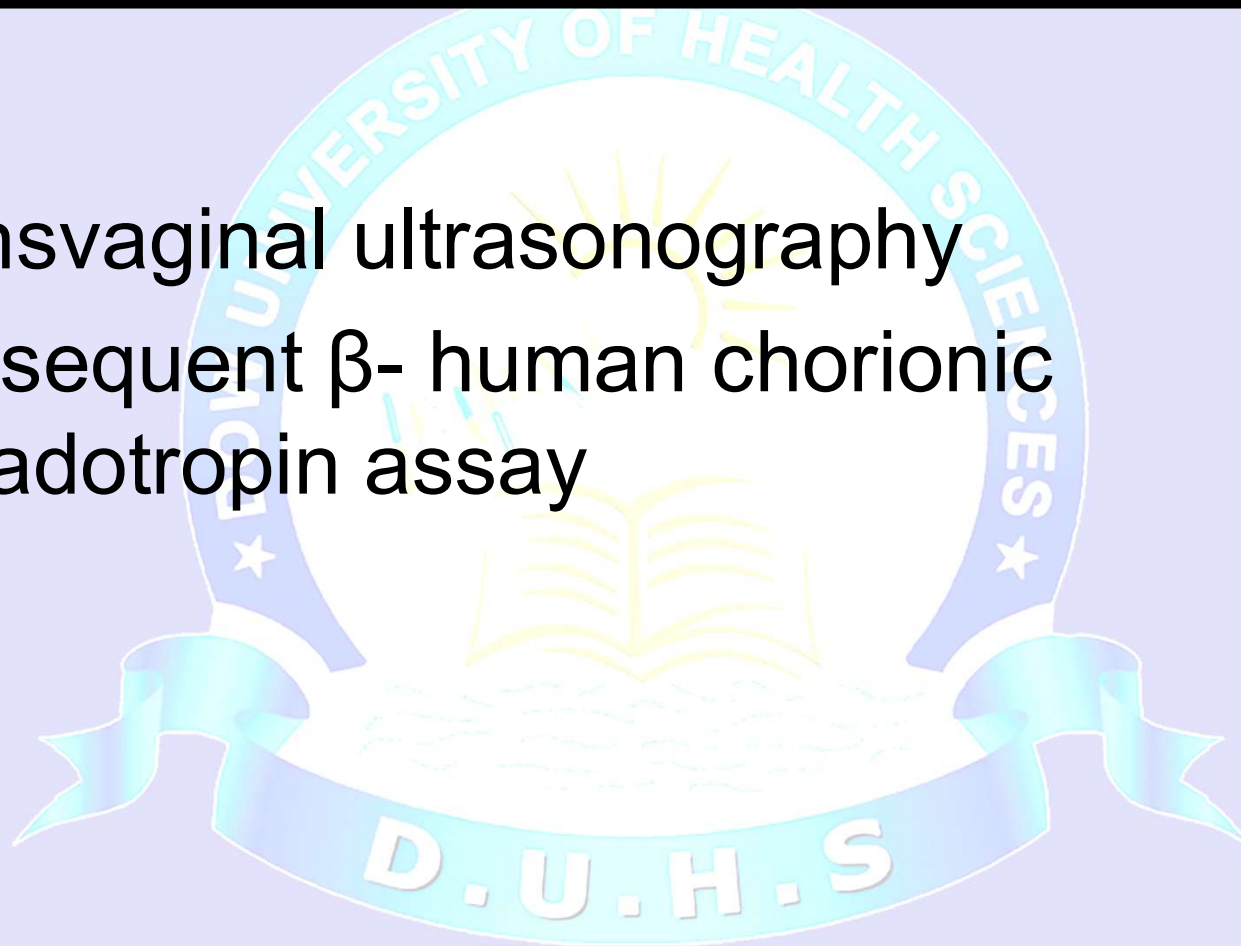
ABNORMAL SITES OF IMPLANTATION (ECTOPIC PREGNANCY)

- Uterin tubes:(95% -98%)
 - » Ampulla
 - » Isthmus
- Abdominal pregnancy (commonly in rectouterine pouch)
- Cervical implantations (unusual)



DIAGNOSTIC TOOLS FOR DETECTION ECTOPIC PREGNANCY

- Transvaginal ultrasonography
- Consequent β - human chorionic gonadotropin assay



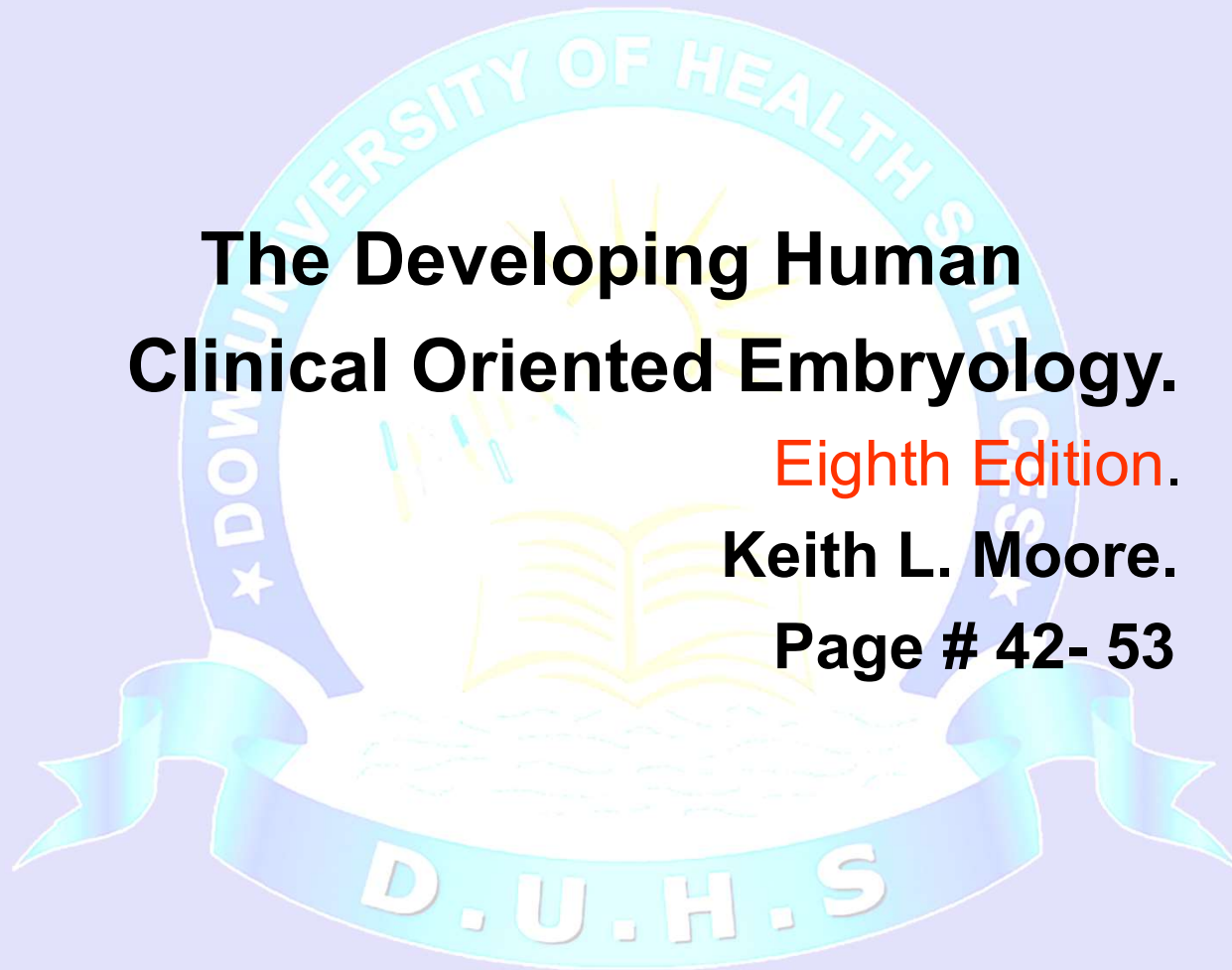
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THANK YOU